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Introduction to R

Msc Molecular Neuroscience:
Gene Expression in the Brain module

Audrys Pauza & Ben Gillard

06/12/23
Bristol

Learning objective

- Show how easy **R** is to use
- Convince you to pick up **R** for your own work
- Not to teach you **R** in one session

Plan

1. Brief introduction
2. Tutorial

WHY USE R?

1. It is completely free to use and Open Source
2. Cross-platform compatible
3. Easy to learn and friendly to use
4. Has a vast and active community
5. Is a complete statistical language
6. Permits advanced data visualisation
7. Gateway to a lucrative career
8. Enables data analysis which is *transparent, shareable and highly-reproducible*

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CRAN
Mirrors
What's new?
Search
CRAN Team

About R
R Homepage
The R Journal

Software
R Sources
R Binaries
Packages

Contributed Packages

Available Packages

Currently, the CRAN package repository features 22838 available packages.

Table of available packages, sorted by date of publication

Table of available packages, sorted by name

CRAN Task Views aim to provide some guidance which packages on CRAN are relevant for tasks related to a certain topic. They provide tools to automatically install all packages from each view. Currently, 48 views are available.

Installation of Packages

Please type `help("INSTALL")` or `help("install.packages")` in R for information on how to install packages from this repository. The manual R Installation and Administration (also contained in the R base sources) explains the process in detail.

Package Check Results

All packages are tested regularly on machines running Debian GNU/Linux, Fedora, macOS (formerly OS X) and Windows.

Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

About Learn Packages Developers Search Get Started

Bioconductor version 3.21 (Release)

Go to 3.22 (Devel)

Find biocViews

- Software (2341)
  - AssayDomain (929)
  - BiologicalQuestion (993)
    - AlternativeSplicing (49)
    - Coverage (161)
    - DifferentialExpression (388)
    - DifferentialMethylation (62)
    - DifferentialPeakCalling (15)
    - DifferentialSplicing (36)
    - DNA3DStructure (12)
    - DriverMutation (1)
    - FunctionalPrediction (34)
    - GeneFusionDetection (2)
    - GenePrediction (27)

Packages found under Software:

Rank based on number of downloads: lower numbers are more frequently downloaded.

Show All entries

Search table:

Package	Maintainer	Title	Rank
BiocVersion	Bioconductor Package Maintainer	Set the appropriate version of Bioconductor packages	1
BiocGenerics	Hervé Pagès	S4 generic functions used in Bioconductor	2
GenomeInfoDb	Hervé Pagès	Utilities for manipulating chromosome names, including modifying them to follow a particular naming style	3
IRanges	Hervé Pagès	Foundation of integer range manipulation in Bioconductor	4
S4Vectors	Hervé Pagès	Foundation of vector-like and list-like containers in Bioconductor	5
XVector	Hervé Pagès	Foundation of external vector representation and manipulation in Bioconductor	6
Biobase	Bioconductor Package Maintainer	Biobase: Base functions for Bioconductor	7
BiocParallel	Martin Morgan	Bioconductor facilities for parallel evaluation	8
Biostrings	Hervé Pagès	Efficient manipulation of biological strings	9

Pretty much limitless applications

Make Music https://jef.works/blog/2021/04/19/randomly-generating-music-with-R/
(from this comprehensive blog: https://jef.works/tags#R)

Call an Uber https://rpubs.com/dbecker7/StrangeRThings

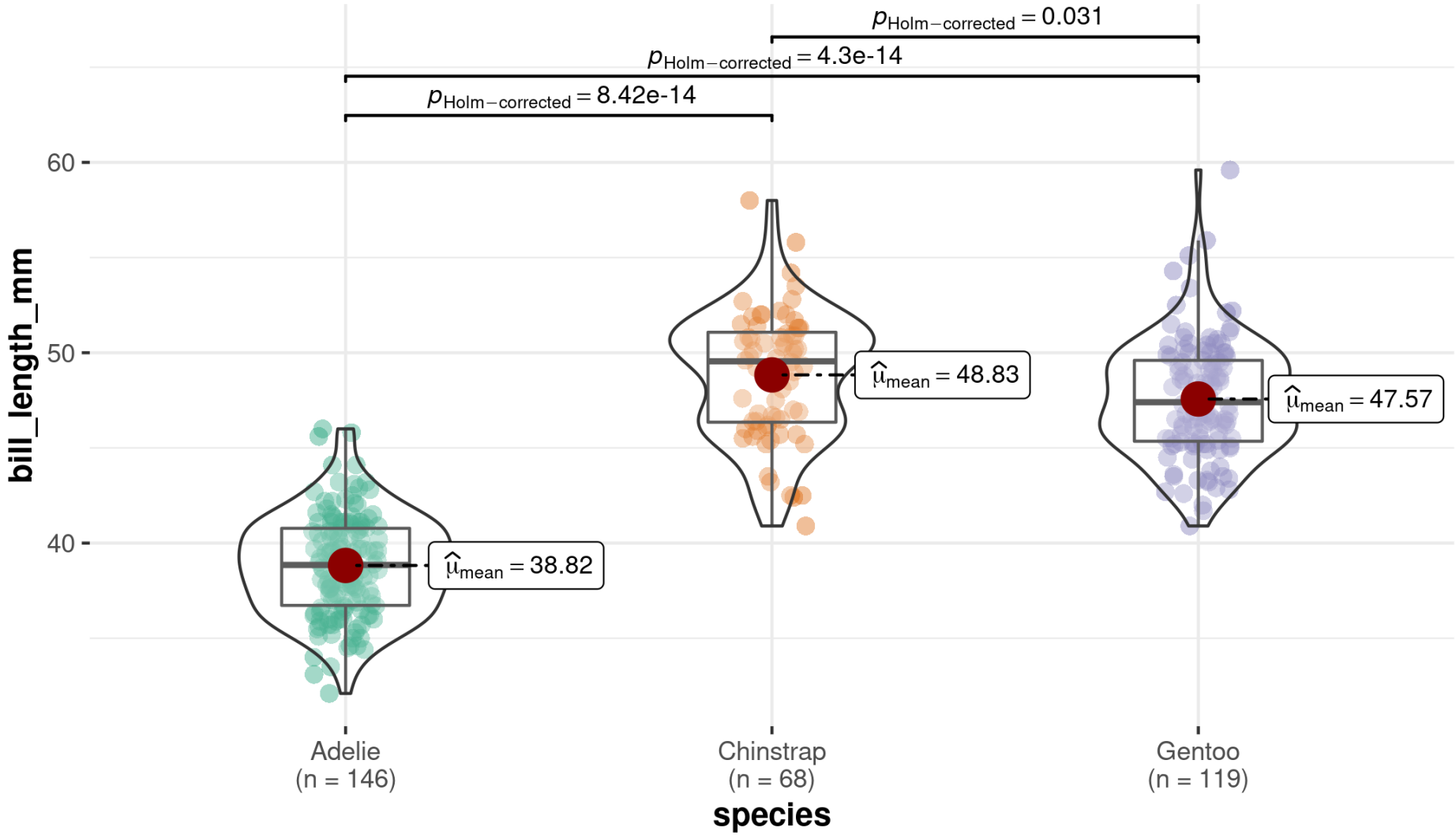
Make animals out of ASCII characters https://sckott.r-universe.dev/articles/cowsay/cowsay.html

Pull a dad joke from the internet https://cran.r-project.org/web/packages/dadjokeapi/dadjokeapi.pdf

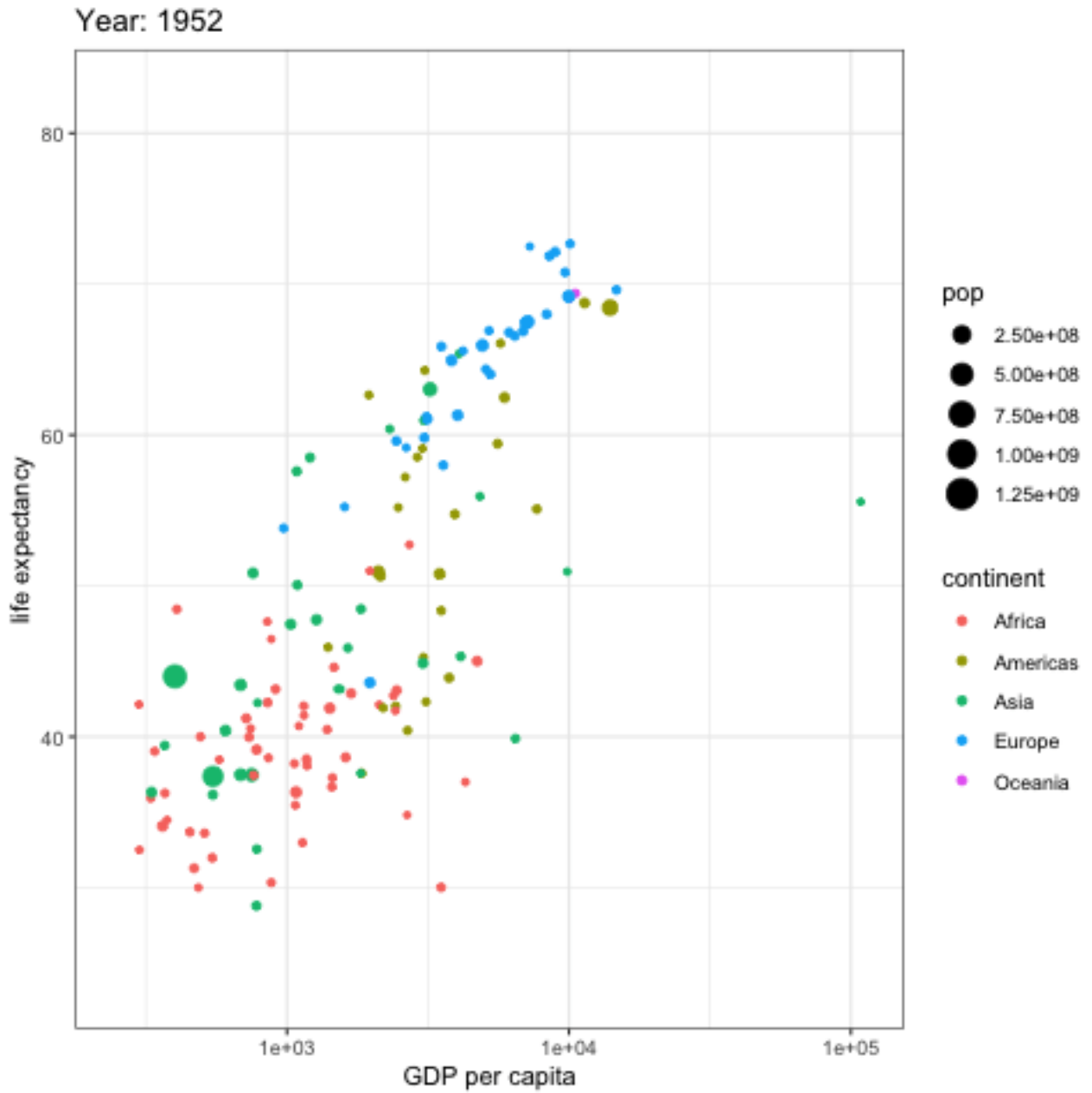
Give yourself an ego boost https://cran.r-project.org/web/packages/praise/praise.pdf

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https://www.r-graph-gallery.com/



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Careers

Growing job market using R
https://careerkarma.com/blog/jobs-that-use-r/

Personal career stories
https://www.nature.com/articles/d41586-023-03120-4

R in industry https://posit.co/resources/videos/the-role-of-r-in-drug-discovery-research-and-development/

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Published: 25 May 2016

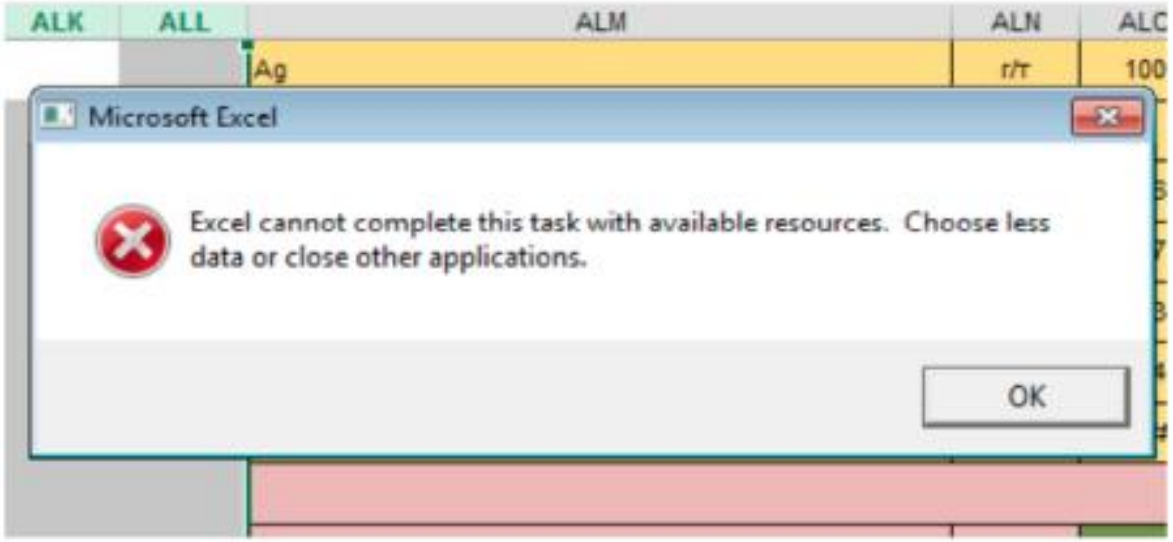
1,500 scientists lift the lid on reproducibility

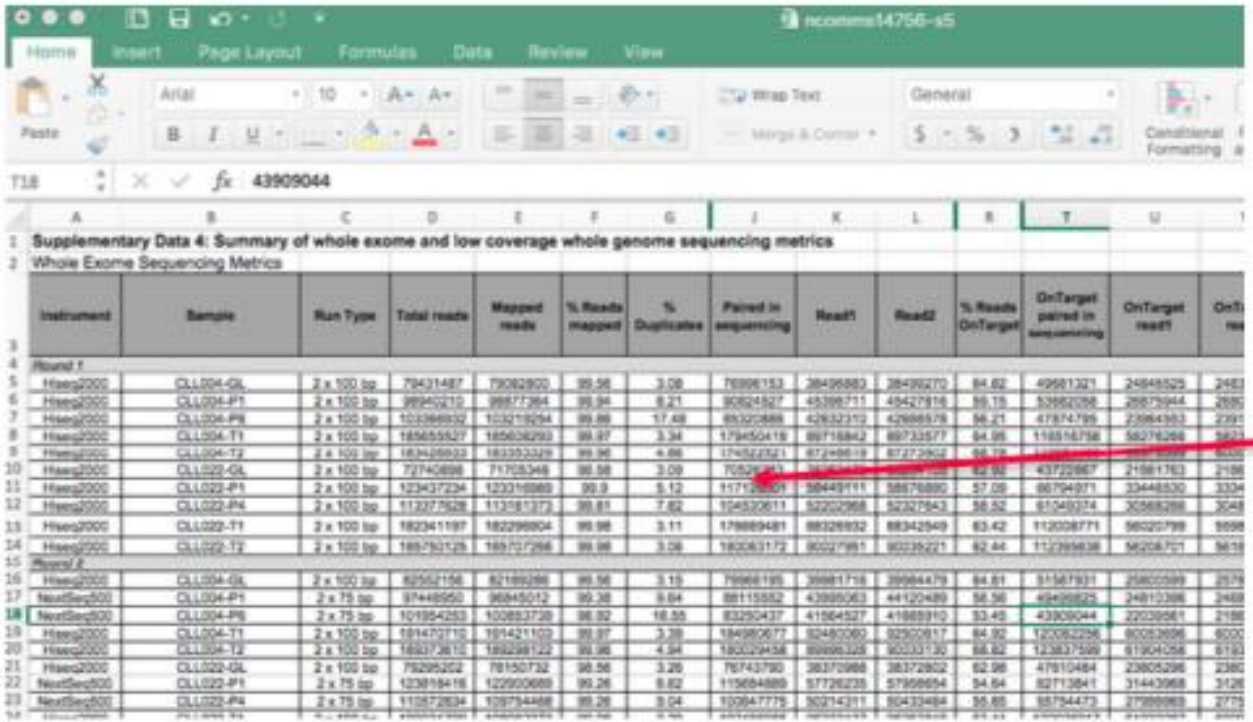
Monya Baker

Nature **533**, 452–454 (2016) | Cite this article

23k Accesses | **1339** Citations | **3877** Altmetric | Metrics







Ziemann et al. Genome Biology (2016) 17:177
DOI 10.1186/s13059-016-1044-7

Genome Biology

COMMENT Open Access

Gene name errors are widespread in the scientific literature

Mark Ziemann[1], Yotam Eren[1,2] and Assam El-Osta[1,3*]

Abstract

The spreadsheet software Microsoft Excel, when used with default settings, is known to convert gene names to dates and floating-point numbers. A programmatic scan of leading genomics journals reveals that approximately one-fifth of papers with supplementary Excel gene lists contain erroneous gene name conversions.

Keywords: Microsoft Excel, Gene symbol, Supplementary data

Abbreviations: GEO, Gene Expression Omnibus; JIF, journal impact factor

frequently reused. Our aim here is to raise awareness of the problem.

We downloaded and screened supplementary files from 18 journals published between 2005 and 2015 using a suite of shell scripts. Excel files (.xls and.xlsx suffixes) were converted to tabular separated files (tsv) with ssconvert (v1.12.9). Each sheet within the Excel file was converted to a separate tsv file. Each column of data in the tsv file was screened for the presence of gene symbols. If the first 20 rows of a column contained five or more gene symbols, then it was suspected to be a list of gene symbols, and then a regular expression (regex) search of the entire column was applied to identify gene symbol errors. Official gene symbols from Ensembl version 82, accessed November 2015, were obtained for

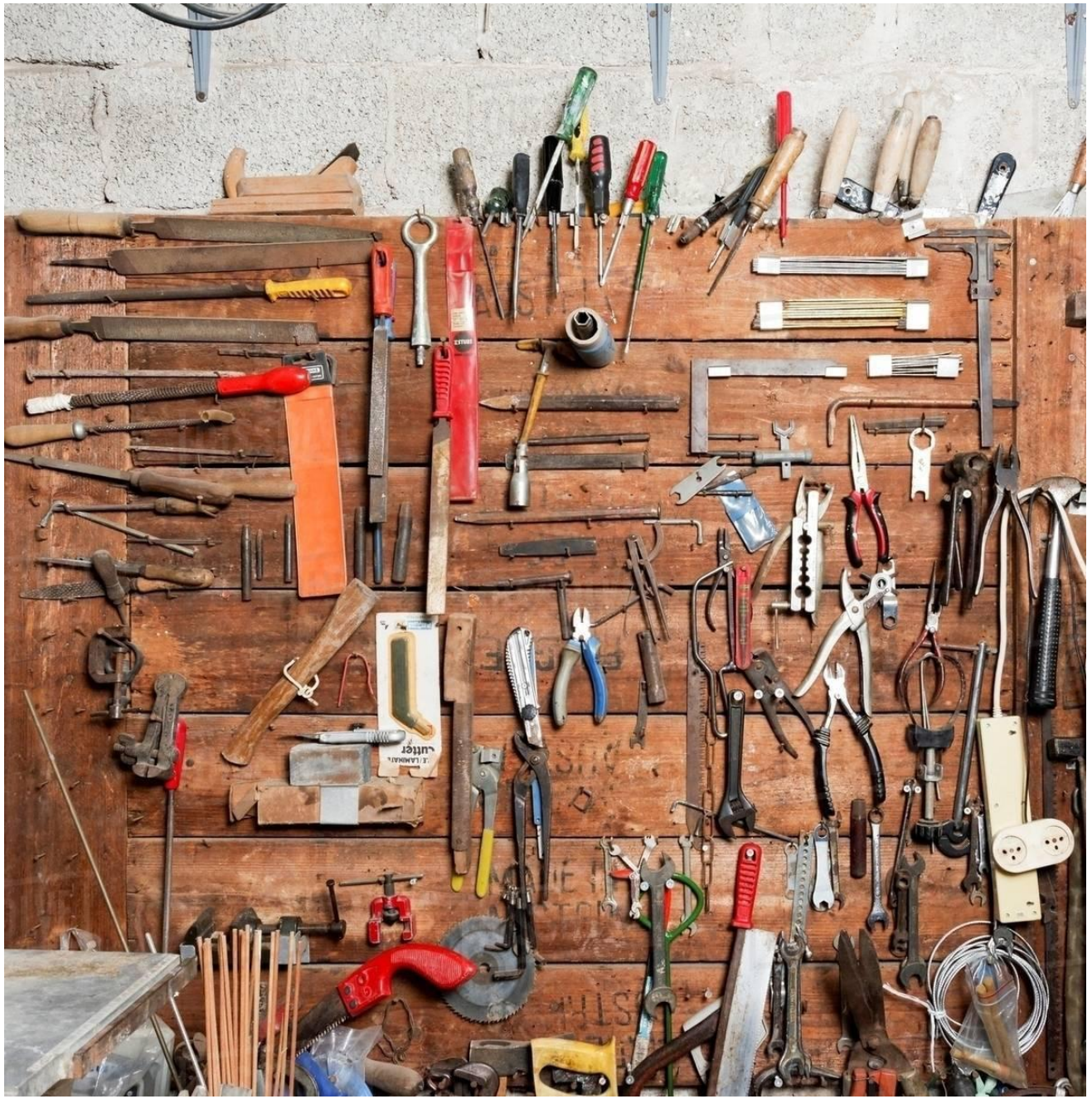
Typo and saved it

TOOLBOX

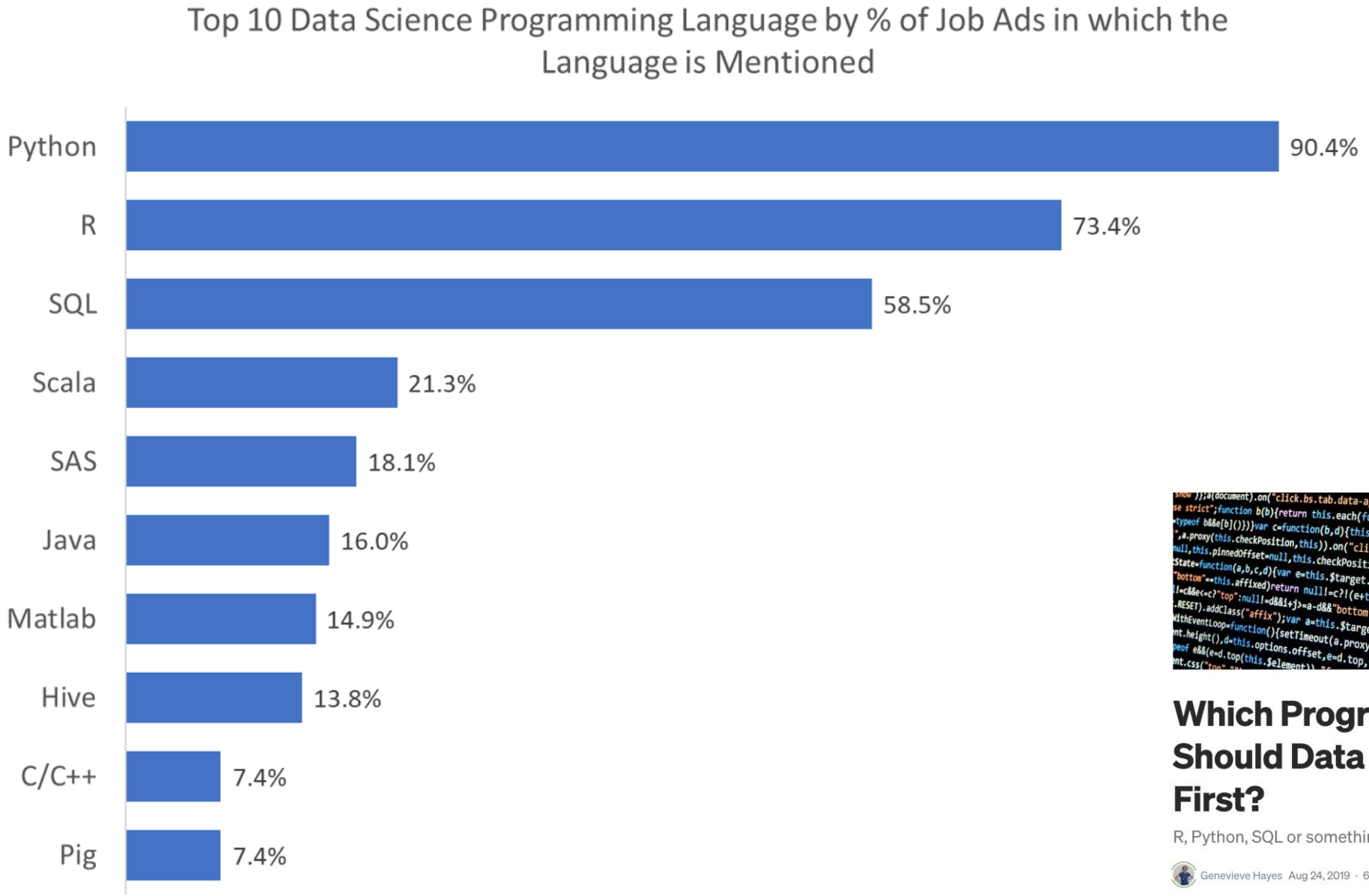
- Operators
- Objects
- Commands
- Functions

Choose the right tool for the job

Bear in mind that you might be **hammering a screw**



Alternatives to R

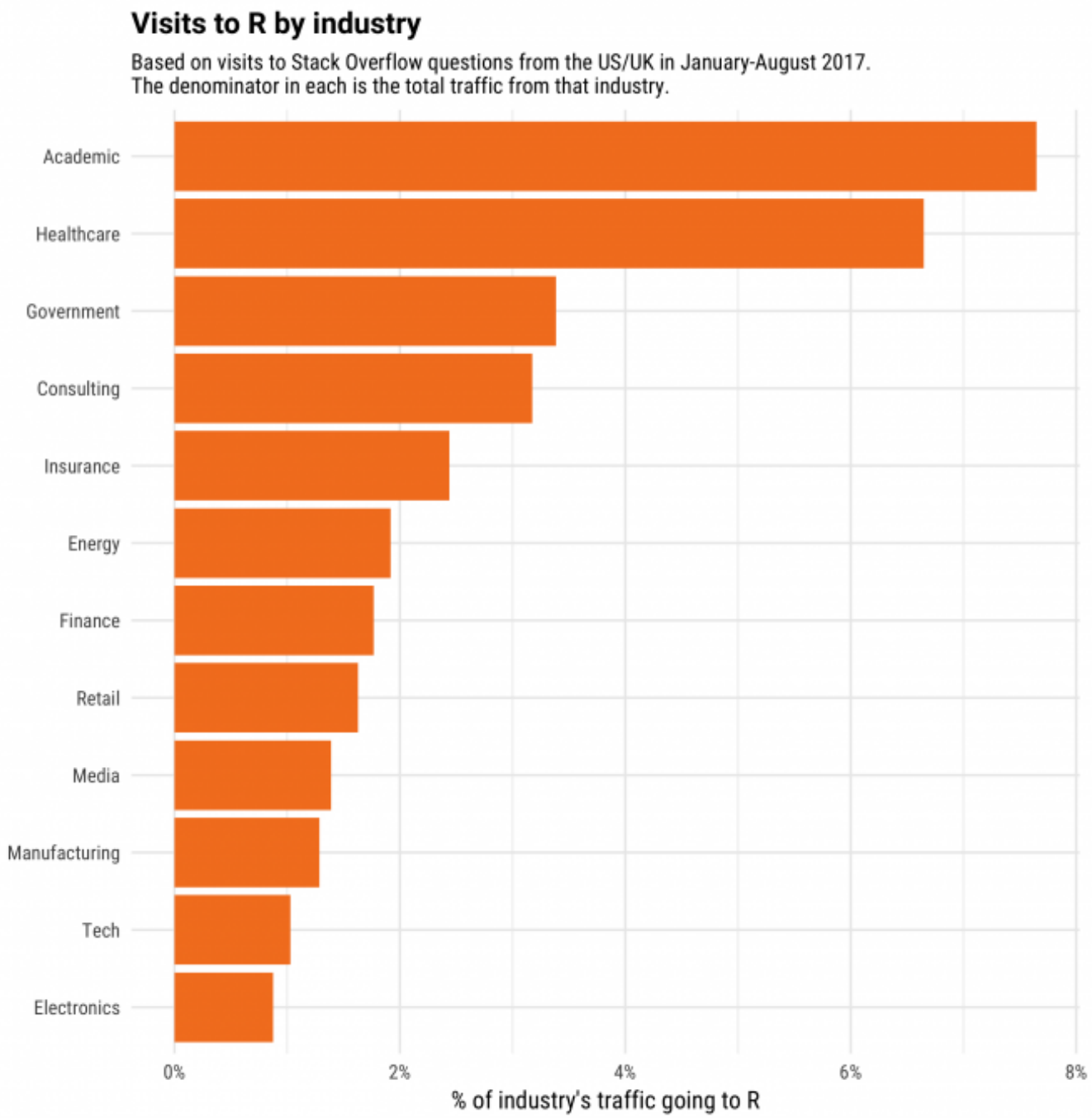




Which Programming Language Should Data Scientists Learn First?

R, Python, SQL or something else?

Genevieve Hayes Aug 24, 2019 · 6 min read ★



https://stackoverflow.blog/2017/10/10/impressive-growth-r/

Can’t I just get my favourite LLM to do it for me?

https://www.nature.com/articles/d41586-026-02715-x

Today’s tutorial - signpost

If it is your first time using R. Go to Intro_to_R_2023_Walkthrough.html. We will go through this together.

****

If you have previous experience with coding. Go to Materials > intro_R.html. Work through this document by yourself or in a group at your own pace. Use troubleshooting section, google your errors, and/or ask each other for help. Come and see Ben if you're still stuck.

****

If you are comfortable with the basics of R. Go to Materials > RNAseqTutorial.html or Materials > ProteomicsTutorial. Work through this document by yourself or in a group at your own pace. Use troubleshooting section, google your errors, and/or ask each other for help. Come and see Ben if you're still stuck.

****

If you just want to learn how to make pretty plots. Go to Materials > ggplot2_Tutorial.html. Work through this document by yourself or in a group at your own pace. Use troubleshooting section, google your errors, and/or ask each other for help. Come and see Ben if you're still stuck.

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R you ready?

We would love to hear your feedback:

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